

StudyMate AI — RAG Test Document

Purpose: This PDF contains selectable, extractable text designed to test PDF extraction, chunking, embedding generation, pgvector storage, semantic retrieval, and document citations in StudyMate AI.

1. Retrieval-Augmented Generation

Retrieval-Augmented Generation, commonly called RAG, combines information retrieval with a language model. Instead of relying only on knowledge already encoded in the model, a RAG system searches an external knowledge source and provides relevant passages to the model before it generates an answer.

A typical RAG pipeline includes document ingestion, text extraction, chunking, embedding generation, vector storage, query embedding, similarity search, context construction, and answer generation. This architecture is useful when an application must answer questions from private, recent, or domain-specific documents.

2. Document Chunking

Large documents should normally be divided into smaller chunks before embedding. Chunking makes retrieval more precise because the system can return only the passages related to the user's question. If chunks are too large, irrelevant information may be included. If chunks are too small, important context may be separated.

StudyMate AI can store each chunk with a document identifier and chunk index. This allows retrieved passages to be traced back to the original document and supports future citation features.

3. Embeddings and Vector Search

An embedding is a numerical representation of text. Semantically similar text tends to produce vectors that are closer together in vector space. StudyMate AI uses the BGE-M3 embedding model through Cloudflare Workers AI. For the current implementation, each embedding contains 1024 numeric dimensions.

The embeddings can be stored in PostgreSQL using the pgvector extension. When a student asks a question, StudyMate AI can create an embedding for the question and compare it with stored document-chunk embeddings. The most similar chunks can then be selected as retrieval context.

4. Semantic Retrieval Example

Suppose a document contains separate sections about machine learning, databases, and computer networks. If the user asks, 'What does an embedding represent?', semantic retrieval should rank the embedding-related section above unrelated sections about networking or SQL.

Semantic retrieval differs from simple keyword matching because it can find conceptually related passages even when the exact wording is different. Retrieval quality can be evaluated by checking whether the expected passages appear among the top results for a set of test questions.

5. Grounding and Citations

A grounded answer should be based on retrieved evidence rather than unsupported model guesses. When evidence is insufficient, the assistant should clearly indicate that the uploaded document does not provide enough information. Document citations can show which file and chunk supported a particular answer.

For a production-quality study assistant, citation correctness matters as much as fluent writing. A citation should point to evidence that actually supports the claim being made.

6. Test Facts for Retrieval

Project codename: Aurora Notebook. This phrase is intentionally distinctive and can be used to test exact retrieval.

Preferred revision interval: 25 minutes of focused study followed by a 5-minute break. This is included only as synthetic test data for retrieval.

Sample course: CS742 Intelligent Information Systems. The fictional final assessment is scheduled for 18 November 2026 and covers vector search, RAG evaluation, and agent workflows.

Important synthetic fact: The Aurora Notebook prototype stores exactly seven retrieval candidates before reranking them to the best three passages. This statement is deliberately specific so that StudyMate AI can be tested with a question such as, 'How many candidates does Aurora Notebook store before reranking?'

7. Suggested Test Questions

After uploading this PDF, useful test questions include: What is RAG? Why are documents chunked? Which embedding model does StudyMate AI use in this test document? How many dimensions are in each embedding? What is the Aurora Notebook codename? How many retrieval candidates are stored before reranking? What should the assistant do when the document does not contain enough evidence?